# Jacopo Umberto Verga, PhD

jacopoumberto.verga@gmail.com | GitHub | LinkedIn | ORCiD | Portfolio


## SUMMARY


**Computational Scientist (Ph.D.)** with 5+ years of experience developing machine learning models and automated workflows for the life sciences. Specialized in multi-omics data integration, GPU-accelerated computing, and generative AI. Experienced in coordinating cross-functional projects and translating complex technical findings for clinical and business stakeholders.


## CORE SKILLS

**Machine Learning & AI**: Generative AI (LLMs, Multi-agent systems, RAG, LangChain, LangGraph, LangSmith), Deep Learning (PyTorch), Transformer Architectures, Statistical Modeling, Predictive Analytics, Clinical ML.
**Bioinformatics & Computational Biology**: Single-cell & Spatial Omics, Multi-omics Integration, Drug Target Identification, Oncology & Hematology Research.
**Software & Infrastructure**: Python, R, Nextflow, Snakemake, Docker, Cloud Platforms (AWS, GCP), High-Performance Computing (HPC), System Benchmarking.
**Project Leadership**: Stakeholder Management, Technical Mentorship, Rapid Prototyping, Cross-functional Collaboration.

## WORK EXPERIENCE

**University of Galway** — Galway, Ireland
*Research scientist - Postdoc* — 06, 2026 – Now

- Developed and deployed end-to-end clinical machine learning models for oncology.
- Architected agentic AI systems to automate research workflows by integrating disparate databases into analytical pipelines.
- Delivered timely clinical biostatistical analyses and technical insights for external partners.
- Developed and instructed genomic data analysis coursework for BSc and MSc students.

**Broad Institute of MIT and Harvard/Dana Farber Cancer Institute** — Boston, MA
*Visiting Researcher, Medical Oncology* — 02, 2024 – 11-2025

- **Led the computational analysis** for multi-modal oncology research, partnering with clinicians to analyse spatial biology data for clinical applications.
- **Managed a 400-sample scRNA-seq study** using UK Biobank data; built GCP-based genomic pipelines with Python and Docker to process high-dimensional datasets.
- **Developed statistical and ML models** for CAR-T patient stratification, supporting myeloma drug discovery and biomarker evaluation - provisional patent.
- **Partnered with StandardBiotools** to integrate 3D imaging and GPU acceleration, reducing pipeline image registration times by 90%.
- **Deployed a U-Net model** for image denoising, optimizing GPU-accelerated training and inference workflows.

**University of Galway** — Galway, Ireland
*Researcher* — 09, 2021 – 11, 2025

- **Analyzed a multi-center ¿522,000-cell dataset** to model immune cell exhaustion, identifying novel therapeutic targets currently in in vivo validation.
- **Developed a computational framework** using maximum-flow network theory and multi-objective genetic algorithms to prioritize key biological drivers.
- **Architected a containerized Nextflow pipeline** automating high-throughput single-cell processing, transcriptional profiling, and cellular interaction inference.
- **Co-developed an AI system** using LLMs, SQL, and RAG to automate unstructured data extraction from literature with tool and code execution.
- **Supervised and mentored 100+ students**; taught genomic data analysis coursework, and cloud computing (AWS).

**CloudCix** — Cork, Ireland
*Tester* — 01, 2023 – 06, 2023

- **Benchmarked deep learning training** across multi-GPU cloud instances using PyTorch and CUDA to evaluate and optimize hardware acceleration.

**Ancona University Hospital** — Ancona, Italy
*Research Assistant, Immunology* — 07, 2019 – 09, 2021

- **Analyzed clinical and genetic datasets** for over 100 immunology patients to evaluate disease risk factors.
- **Conducted safety and efficacy analyses** for CSL Behring intravenous pharmaceutical products.
- **Collaborated with clinical teams** to translate complex biological datasets into research insights.

**Queen's University Belfast** — Belfast, UK
*Intern* — 02, 2020 – 08, 2020

- **Executed virtual screenings of 500+ small molecules**, utilizing molecular dynamics and protein-protein interaction networks.

## EDUCATION

**University of Galway** — Galway, Ireland
*PhD, Genomics Data Science* — 09, 2021 – 11, 2025
**Marche Polytechnic University** — Ancona, Italy
*Master's Degree, Molecular and Applied Biology* — 2018 – 2021
**Grade:** 110L/100 (4.0 GPA)
**Marche Polytechnic University** — Ancona, Italy
*Bachelor's Degree, Biological Sciences* — 2015 – 2018
**Grade:** 108/100 (3.9 GPA)

## CERTIFICATIONS

**Anthropic:** Model Context Protocol (MCP), Agents Skills (**ongoing**)
**Google:** GenAI Intensive Course (**04/2025**)
**NVIDIA:** Building Transformer-based NLP Applications (**11/2024**)
**NVIDIA:** Fundamentals of Deep Learning (**10/2024**)
**CITI Program:** Good Clinical Practice (GCP) for Clinical Research (**02/2024**)
**CITI Program:** Human Research (**02/2024**)
**University of Galway:** Project Management in the Real World (**11/2024**)

## AWARDS

**Thomas Crawford Hayes Travel Fund** winner, **€2300**.
**Longhack & VitaDAO 2023 hackathon** Winning team, Team Leader, **€1000 + 1000 VITA-Coin**
**1$^{st}$ prize for the Decisive Data Podium Session** at the CSE Research Day, **€200**

## PROJECTS

**FinanceBro,** investing wallet simulator, developed with AI-coding assistant crew (OpenCode, Agents, Skills, LLMs).
**miRKatAI: Agentic AI for miRNA research,** Google Gen AI, SQL, Prompt engineering, GroundSearch, Streamlit
**Breast Cancer Chatbot: Assistant to analyze clinical/molecular data,** Bioinformatics Workflows, AI Agents, LLM, Autoencoders
**AI Maria: Fine-tuned LLM for Myeloma Research Assistant,** Transformers, Fine tuning, RoBERTa

## LANGUAGES

**Language:** English (C1 - Full Professional Proficiency), Italian (Native), Spanish (B2)

## PUBLICATIONS


[1] Karen Guerrero-Vazquez et al. "miRKatAI: An Integrated Database and Multi-agent AI system for microRNA Research". In: (2025). DOI: 10.48550/ARXIV.2508.08331. URL: https://arxiv.org/abs/2508.08331.

[2] **Verga, Jacopo U**. and Foster, Kane and Cordas Dos Santos, David M. et al. "Bone Marrow Spatial Signatures Predict Survival Outcomes and Toxicities after CAR-T Therapy in Patients with Relapsed/Refractory Multiple Myeloma". In: *Nature Medicine - Under Review* (2026).

[3] Yoshinobu Konishi et al. "Bone Marrow Spatial Signatures Predict Survival Outcomes and Toxicities after CAR-T Therapy in Patients with Relapsed/Refractory Multiple Myeloma". In: *Blood* 144.Supplement 1 (Nov. 2024), pp. 590–590. ISSN: 1528-0020. DOI: 10.1182/blood-2024-204144. URL: http://dx.doi.org/10.1182/blood-2024-204144.

[4] **Verga, Jacopo U** et al. "NK Cell States in Multiple Myeloma: Pathways to Novel Immunotherapies". In: *Blood* 144.Supplement 1 (Nov. 2024), pp. 6857–6857. ISSN: 1528-0020. DOI: 10.1182/blood-2024-210297. URL: http://dx.doi.org/10.1182/blood-2024-210297.

[5] David Cordas dos Santos et al. "P-203 Ability To Perform Spatial Transcriptomics in FFPE Decalcified Bone Marrow Samples of Patients With Precursor Myeloma". In: *Clinical Lymphoma Myeloma and Leukemia* 24 (Sept. 2024), S154–S155. ISSN: 2152-2650. DOI: 10.1016/s2152-2650(24)02106-2. URL: http://dx.doi.org/10.1016/S2152-2650(24)02106-2.

[6] **Verga, Jacopo U** et al. "Genomic technology advances and the promise for precision medicine". In: *Therapeutic Drug Monitoring*. Elsevier, 2024, pp. 355–371. ISBN: 9780443186493. DOI: 10.1016/b978-0-443-18649-3.00007-0. URL: http://dx.doi.org/10.1016/B978-0-443-18649-3.00007-0.

[7] **Verga, Jacopo U** et al. "A Systems Biology Approach Reveals the Endocrine Disrupting Potential of Aflatoxin B1". In: *Exposure and Health* 16.2 (May 2023), pp. 321–340. ISSN: 2451-9685. DOI: 10.1007/s12403-023-00557-w. URL: http://dx.doi.org/10.1007/s12403-023-00557-w.

[8] Zachary Boswell et al. "In-Silico Approaches for the Screening and Discovery of Broad-Spectrum Marine Natural Product Antiviral Agents Against Coronaviruses". In: *Infection and Drug Resistance* Volume 16 (Apr. 2023), pp. 2321–2338. ISSN: 1178-6973. DOI: 10.2147/idr.s395203. URL: http://dx.doi.org/10.2147/IDR.S395203.

[9] Maria Giovanna Danieli et al. "Common Variable Immunodeficiency in Elderly Patients: A Long-Term Clinical Experience". In: *Biomedicines* 10.3 (Mar. 2022), p. 635. ISSN: 2227-9059. DOI: 10.3390/biomedicines10030635. URL: http://dx.doi.org/10.3390/biomedicines10030635.

[10] **Verga, Jacopo U** et al. "Integrated Genomic and Bioinformatics Approaches to Identify Molecular Links between Endocrine Disruptors and Adverse Outcomes". In: *International Journal of Environmental Research and Public Health* 19.1 (Jan. 2022), p. 574. ISSN: 1660-4601. DOI: 10.3390/ijerph19010574. URL: http://dx.doi.org/10.3390/ijerph19010574.

[11] George S. Hanna et al. "Contemporary Approaches to the Discovery and Development of Broad-Spectrum Natural Product Prototypes for the Control of Coronaviruses". In: *Journal of Natural Products* 84.11 (Oct. 2021), pp. 3001–3007. ISSN: 1520-6025. DOI: 10.1021/acs.jnatprod.1c00625. URL: http://dx.doi.org/10.1021/acs.jnatprod.1c00625.

[12] Veronica Pedini et al. "Incidence of malignancy in patients with common variable immunodeficiency according to therapeutic delay: an Italian retrospective, monocentric cohort study". In: *Allergy, Asthma & Clinical Immunology* 16.1 (June 2020). ISSN: 1710-1492. DOI: 10.1186/s13223-020-00451-z. URL: http://dx.doi.org/10.1186/s13223-020-00451-z.

[13] Maria Giovanna Danieli et al. "A Case of CVID-Associated Inflammatory Bowel Disease with CTLA-4 Mutation Treated with Abatacept". In: *Archives of Clinical and Medical Case Reports* 03.06 (2019). ISSN: 2575-9655. DOI: 10.26502/acmcr.96550124. URL: http://dx.doi.org/10.26502/acmcr.96550124.